

1. Administrative & Study Identification

Full Study Title: KEYMAKER-U01 Substudy 01E: A Phase 2 Umbrella Study With Rolling Arms of Investigational Agents With or Without Chemotherapy in Combination With Pembrolizumab in Treatment of Participants With Newly Diagnosed Resectable Stages II-IIIB (N2) Non-small Cell Lung Cancer (NSCLC) **Short Title/Acronym:** Pembrolizumab (MK-3475) Plus Investigational Agents in Resectable Non-small Cell Lung Cancer (NSCLC) (MK-3475-01E/KEYMAKER-U01) **Protocol Number:** MK-3475-01E (3475-01E) **NCT Number:** NCT06788912 **Posting ID:** 906228081365110469 **Sponsor/Company Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475), Sacituzumab tirumotecan (MK-2870/SKB264), and Chemotherapy (Cisplatin, Gemcitabine, Pemetrexed, Carboplatin, Paclitaxel) **Phase of Development:** Phase 2 (Phase ID: PHASE2) **Phase Requirement:** Trial must be in PHASE2 (Criteria Type: PHASE_REQUIREMENT) **Trial Status:** RECRUITING (Status Description: Currently recruiting participants) **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: The main goals are after treatment given before surgery, to measure the number of people who have no signs of cancer cells in tumors and lymph nodes removed during surgery; and to learn about whether the cancer gets smaller or goes away by measuring the number of people with a certain number of living cancer cells in the tumor removed during surgery. **Secondary Objectives:** To evaluate Event-free Survival (EFS), the percentage of participants who report at least 1 Adverse Event (AE), the percentage of participants who discontinue study treatment due to an AE, and long-term Overall Survival (OS).

2.2 Background & Rationale To address the unmet need to improve outcomes in early-stage NSCLC by investigating the addition of multiple investigational agents (such as the antibody-drug conjugate sacituzumab tirumotecan) with or without chemotherapy to pembrolizumab. The approach evaluates whether combining immunotherapy with standard or novel therapies before surgery (neoadjuvant) and continuing immunotherapy after surgery (adjuvant) can increase the number of patients achieving a pathological complete response and improve survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study with Rolling Arms) **Control Type:** Active Comparator (Reference arm: Pembrolizumab + Chemotherapy) **Blinding:** Open-label **Randomization:** 1:1, Stratified by histology (squamous vs. nonsquamous) and tumor stage (II vs. III).

3.2 Planned Enrollment & Duration Targeted Enrollment: 60 participants **Number of Sites:** 34 sites globally **Estimated Study Start:** 20-Mar-2025 **Estimated Primary Completion:** 06-Feb-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has previously untreated and pathologically confirmed resectable Stage II, IIIA, or IIIB (N2) non-small cell lung cancer (NSCLC). 2. Able to undergo protocol therapy, including necessary surgery. 3. Confirmation that epidermal growth factor receptor (EGFR) -directed therapy is not indicated as primary therapy. 4. Has an Eastern Cooperative Oncology Group (ECOG) performance status of either 0 or 1 as assessed within 10 days before initiation of study intervention. 5. Is able to provide archival or newly obtained core/excisional biopsy of the primary lung tumor or lymph node metastasis.

4.2 Exclusion Criteria 1. Has one of the following tumor locations/types: NSCLC involving the superior sulcus, large-cell neuro-endocrine cancer, mixed tumors containing small cell and non-small cell elements, or sarcomatoid tumor. 2. Has Grade ≥ 2 peripheral neuropathy. 3. Has history of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 4. Has active inflammatory bowel disease requiring immunosuppressive medication or previous history of inflammatory bowel disease (eg, Crohn's disease, ulcerative colitis, or chronic diarrhea). 5. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 6. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 7. Received prior radiotherapy within 2 weeks of start of study intervention, or radiation related toxicities, requiring corticosteroids. 8. Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any other form of immunosuppressive therapy within 7 days prior to the first dose of study intervention. 9. Known additional malignancy that is progressing or has required active treatment within the past 5 years. 10. Severe hypersensitivity ($\geq$ Grade 3) to pembrolizumab and/or any of its excipients. 11. Active autoimmune disease that has required systemic treatment in the past 2 years. Replacement therapy (e.g. thyroxine, insulin, or physiologic corticosteroid) is allowed. 12. History of (noninfectious) pneumonitis/interstitial lung disease that

required steroids or has current pneumonitis/interstitial lung disease. 13. Active infection requiring systemic therapy. 14. Hepatitis B (defined as hepatitis B surface antigen [HBsAg] reactive) or Hepatitis C virus (defined as detectable hepatitis C virus (HCV) ribonucleic acid (RNA) [qualitative]) infection. 15. Known history of human immunodeficiency virus (HIV) infection. 16. History of allogeneic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm 1 (Reference Arm):** Neoadjuvant therapy of 4 cycles of pembrolizumab 200 mg IV + chemotherapy IV Q3W. * **Arm 2 (Treatment Arm):** Neoadjuvant therapy of 4 cycles of pembrolizumab 200 mg IV Q3W + 6 cycles of sacituzumab tirumotecan 4 mg/kg IV Q2W. * **Adjuvant Phase (Both Arms):** Following surgery, up to 13 cycles of adjuvant pembrolizumab 200 mg IV Q3W.

Route of Administration: Intravenous (IV). **Duration of Treatment:** Up to 4-6 cycles of neoadjuvant therapy prior to surgery, followed by up to 13 cycles (approx. 39 weeks) of adjuvant therapy.

5.2 Concomitant & Prohibited Therapy Permitted: Paracetamol (Preferred Name: paracetamol, ATC Code: N02BE01) is allowed for symptomatic relief and pain management (refer to Appendix A for all permitted trade names). Replacement therapy (e.g., thyroxine, insulin, or physiologic corticosteroid). Optional adjuvant platinum-based doublet chemotherapy up to 4 cycles may be given at the investigator's discretion. **Prohibited:** Live or live-attenuated vaccine within 30 days before the first dose; systemic steroid therapy (dosing exceeding 10 mg daily of prednisone equivalent) or any other form of immunosuppressive therapy within 7 days prior to intervention.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Tumor biopsy for biomarker analysis, Baseline Imaging
Neoadjuvant	Cycles 1-4/6	Pembrolizumab + Chemotherapy OR Pembrolizumab + Sacituzumab tirumotecan IV	
Surgery	Post-neoadjuvant	Tumor resection and pathological assessment for pCR / %RVT	Adjuvant Cycles 1-13 Adjuvant Pembrolizumab 200 mg Q3W
End of Study	Follow-up	Survival Follow-up, EFS, Imaging (≤4 weeks before adjuvant start), Safety Assessments	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Pathological Complete Response (pCR): pCR is defined as absence of residual viable invasive cancer on hematoxylin- and eosin-stained slides of the resected lung specimen and lymph nodes. **2. Percent Residual Viable Tumor (%RVT):** %RVT is defined as the percentage of residual tumor estimated by comparing the estimated cross-sectional area of viable tumor with estimated cross-sectional areas of remainder of tumor bed. The tumor bed is defined as the area of tissue occupied by viable tumor or tumoral regression (includes areas of necrosis, foamy macrophages, giant cell reaction, cholesterol cleft granuloma, and inflammation).

7.2 Secondary & Exploratory Endpoints * Percentage of Participants Who Report at Least 1 Adverse Event (AE). * Percentage of Participants Who Discontinue Study Treatment Due to an AE. * Event-free Survival (EFS) per investigator review. * Overall Survival (OS) and Distant Metastasis-Free Survival (DMFS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine safety monitoring at each visit; AEs will be graded per NCI CTCAE v5.0. **Serious Adverse Events (SAEs):** Specific monitoring for perioperative surgical complications, pneumonitis, and severe infusion reactions. **Data Monitoring Committee (DMC):** External/Internal monitoring safety reviews as per standard Merck protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all treatment-emergent adverse events. **Sample Size Justification:** A targeted enrollment of 60 patients (randomized 1:1) will provide sufficient data to evaluate early efficacy signals (pCR and %RVT) for the substudy rolling arms. **Handling of Missing Data:** Standard imputation methods for time-to-event endpoints (EFS/OS) and handling of drop-outs per study data analysis plans.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard global multi-center onsite and remote monitoring. **Audit Trail:** Verified electronic Case Report Forms (eCRF), RECIST v1.1 compliance, and independent pathology audits for tissue samples.

11. Study Identifiers & Governance

Primary ID: MK-3475-01E **Registry Number:** NCT06788912 **Posting ID:** 906228081365110469 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** Pembrolizumab (MK-3475) Plus Investigational Agents in Resectable Non-small Cell Lung Cancer (NSCLC) (MK-3475-01E/KEYMAKER-U01) **Official Regulatory Title:** KEYMAKER-U01 Substudy 01E: A Phase 2 Umbrella Study With Rolling Arms of Investigational Agents With or Without Chemotherapy in Combination With Pembrolizumab in Treatment of Participants With Newly Diagnosed Resectable Stages II-IIIB (N2) Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework (NSCLC Specific)

Condition Name: Lung Cancer **Disease Areas:** Lung Neoplasm
Maligant Preferred UMLS Name: Non-Small Cell Lung Carcinoma
Preferred Name: Malignant neoplasm of lung **Semantic Type:** Neoplastic Process **Definition:** A primary or metastatic malignant neoplasm involving the lung. **Classification Codes:** * **MeSH Code:** D008175 * **HPO Code:** HP:0100526 * **SNOMED ID:** 363358000 **SNOMED Hierarchy:** * 430621000 | Malignant neoplasm of lower respiratory tract | Malignant neoplasm of lower respiratory tract (disorder) * 188361007 | Malignant neoplasm of thorax | Malignant neoplasm of thorax (disorder) * 126713003 | Neoplasm of lung | Neoplasm of lung (disorder) * 39607008 | Lung structure | Lung structure (body structure) * 1240414004 | Malignant neoplasm | Malignant neoplasm (morphologic abnormality) * 64572001 | Disease | Disease (disorder)

Diagnosis Threshold: Pathologically confirmed resectable Stage II, IIIA, or IIIB (N2) NSCLC without EGFR mutations. **Data Source Level:** 5

13. Study Procedures & Methodology

Management Pathway: Neoadjuvant systemic treatment, followed by surgical resection, and adjuvant systemic treatment pathway
Education Protocol (HCP): Clinical protocol, safety management, and RECIST v1.1 evaluation training **Education Protocol (Patient):** Informed consent, treatment administration education, and surgical preparation **Quality Audit Mechanism:** Blinded independent pathology review (BIPR) for primary endpoints

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 7 years (Primary Completion: Feb 2032) **Onsite Visit Cadence:** Q2W/Q3W during neoadjuvant phase,

Q3W during adjuvant therapy **Remote Monitoring Frequency:** Routine survival follow-up periodic contacts post-treatment **Checklist Review Interval:** End of each treatment cycle (every 3 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						

Appendix A: Permitted Concomitant Medications (Paracetamol Trade Names)

To manage symptoms such as mild pain or fever, **Paracetamol (Preferred Name: paracetamol, ATC Code: N02BE01)** may be administered. The following trade names are registered in the study framework as permissible variants:

Syncol, Orbivan CF, Duraxin, Mejoralito, Anacin PM Aspirin Free, Bayer Select Night Time Pain, Sominex Pain Relief, EX-Pain Sleep, Leg Ease, XL-DOL, Ringl, Un-Aspirin, Bayer Migraine, Arthriten Inflammatory Pain, Sedalmex, Primlev, Orbivan, Counteract PM, Little Fevers, Panadol PM, Pharbetol, Cetazone, CounterAct Pain, Panadol Menstrual Relief, Capacet, Nortemp, Bactimicina, Children's Dimetapp Fever and Sore Throat, Zolvit, Arthriten, BF-Paradac, Acetadriak, Ofirmev, Backaid, Pamprin Max Formula, Percogesic Reformulated Jan 2011, Pamprin Cramp Formula, Pamprin Multi-Symptom, Cetafen, Exaprin, Painaid, Painaid ESF, Painaid BRF, Acetadryl, Pyregesic, Tylagesic, Tactinal, Zflex, Xolox, Primalev, Zamicet, Cocet, BP Poly-650, Backprin, Prodrin, Liquicet, Flextra Plus, Magnacet, Zgesic, Maxiflu CD, Cotabflu, Excedrin Quick Tab, Lagesic, Alpain, Propacet, Migran-A, Zerlor, Codrix, Vistra, Cafgesic, Comtrex Sore Throat Relief, Excedrin Back & Body, Anacin Advanced Headache Formula, Trezix, Dolgic Plus, Pediaphen, Fortolin, MigraTen, Panlor DC Reformulated Jan 2008, Combiflex ES, Perloxx, Narvox, QFlex, Durabac Forte, By Ache, Dolorex brand of acetaminophen-phenyltoloxamine, Lorcet, Elixsure Fever/Pain, Excedrin Tension Headache, Epidrin, Staflex, Valorin Extra, Hycet, Levacet, Ed Flex, Anabar, Quala Cet, Lobac, Dolgic LQ, Durabac, Darvocet, Migralam, Migrend, Taxadone, Frenadol, Phenco-Care, Acuflex, Dologesic, Ali Flex, Xodol, Hyflex, Vopac, Flextra, Vidone, Tylefon, Dolorex Forte, Balacet, Novagesic, Panasorb, Lopap brand of acetaminophen, Tylenol Severe Allergy, Contragesic, Nonbac, Procet, Promacet, Panlor SS, Phrenilin with Caffeine and Codeine, Relagesic, Ultracet, Va-zone, Prolate, Butex Forte, Maxidone, Womens Tylenol Menstrual Relief, Rhinoflex, Ex-Pain, Acephen, Calpol, Fanalgic, Infadrops, Panadol Extra, Paracets, Panaleve, Paramol, Placidex, Salzone, Tylox, Paldesic, Disprol Junior, Calpol Six Plus, Medinol Over Six, Medinol Pediatric, Medinol Under Six, Tylenol, Panadol,

Triad, Combiflex, Fioricet, Aceta, Aceta-Gesic, Actamin, Adprin B, Allay, Alidrin, Alagesic, Americet, Altenol, Amidrine, Anacin AF, Anacin-3 Maximum Strength, Anolor, Anexsia, Anoquan, Anolor DH5, Apacet, Apagesic, Apapgesic, Apra, Apra, Arcet, Axocet, Bancap HC, Bayer Select, Bromo Seltzer, Bucet, Bupap, Capital and Codeine, Cephadyn, Ceta Plus, Child Silapap, Childrens Panadol, Co-Gesic, Conacetol, Colrex, Dolacet, Dolgic, Dolagesic, Dolmar, Dolono Infant, Dolorex, Dolono, Duogesic, Duradrin, EZ III, Ed-APAP, Endocet, Endolor, Esgic, Ezol, Excedrin, Excedrin PM, Femcet, Feverall, Fioricet with Codeine, Fiorpap, G-1, Genace brand of acetaminophen/aspirin/caffeine, Genapap PM, Genasec, Genapap Extra Strength, Genebs, Genapap, Gericet, Geone, Halenol, Hyco-pap, Hycomed, Hy-Phen, Hycomine Compound, Hydrocet, Hydrogesic, Ide-cet, I.D.A., Infantaire, Isocet, Iso-Acetazone, Isocom, Legatrin PM, Liquiprin, Lurline PMS, Major-gesic, Mapap Junior Strength, Mapap, Mapap PM, Margesic, Margesic-H, Margesic #3, Masophen PM, Mardol, Marten-Tab, Masophen, Meda Cap, Medigesic, Medipain 5, Migquin, Migrapap, Migratine, Migrex, Midchlor, Migrazone, Mitride, Minotal, Mygracet, Neopap Suppettes, Norco, Oncet, Pain-Eze, Pain-Gesic, Panacet, Panex 500, Pacaps, Panex, Panritis Forte, Panlor, Padiapap, Perisine, Pharmagesic, Phenagesic, Phenaphen with Codeine, Phenylgesic, Phrenilin, Phrenilin Forte, Polygesic, Prominol, Q-Nol, Q-Gesic, Q-Nol Extra Strength, Redutemp, Repan, Repan CF, Rid-A-Pain, Roxicet, Ridenol, Roxilox, S-T Febrol, Saleto, Sedapap, St. Joseph Aspirin-Free, Stagesic, Supac, T-Gesic, T-Panol, T-Painol, Tempra 1, Tempra 2, Tempra Quicklets, Tempra 3, Tencet, Tenake, Tencon, Triaprin, Two-Dyne, Tylophen, Tylenol PM, Tycolene, Tylenol Gelcap Extra Strength, Tylenol with Codeine, Uni-Perr, Ugesic, Uni-Ace, Unisom with Pain Relief, Uniserts, Uni-Ace Child, Uni-Case, Vanacet, Vanquish, Vendone, Vitoxapap, Vitapap, XS Pain Reliever, XS No Aspirin PR, Wygesic, Zebutal, Zydone, Vicodin, Lortab, DuoCet, Premsyn PMS, Talacen, Tempra, Percogesic, Percocet, Midrin, Combogesic, Vtol, Dvorah, Apadaz, Arthriten Max, Nalocet, Femadrine, Dolofin, Sed-Max, Panlor Reformulated Jan 2018, Zyfrel, Excedrin Mild Headache, Excedrin PM Triple Action, Ninjacof A, Allzital, Dologesic Reformulated Jun 2016, Vanatol, Nanomol suspension, Backaid IPF, Xartemis, Lynox, Dolofin Infantil, Nodolor, Uricalm Intensive, Verdrocet.